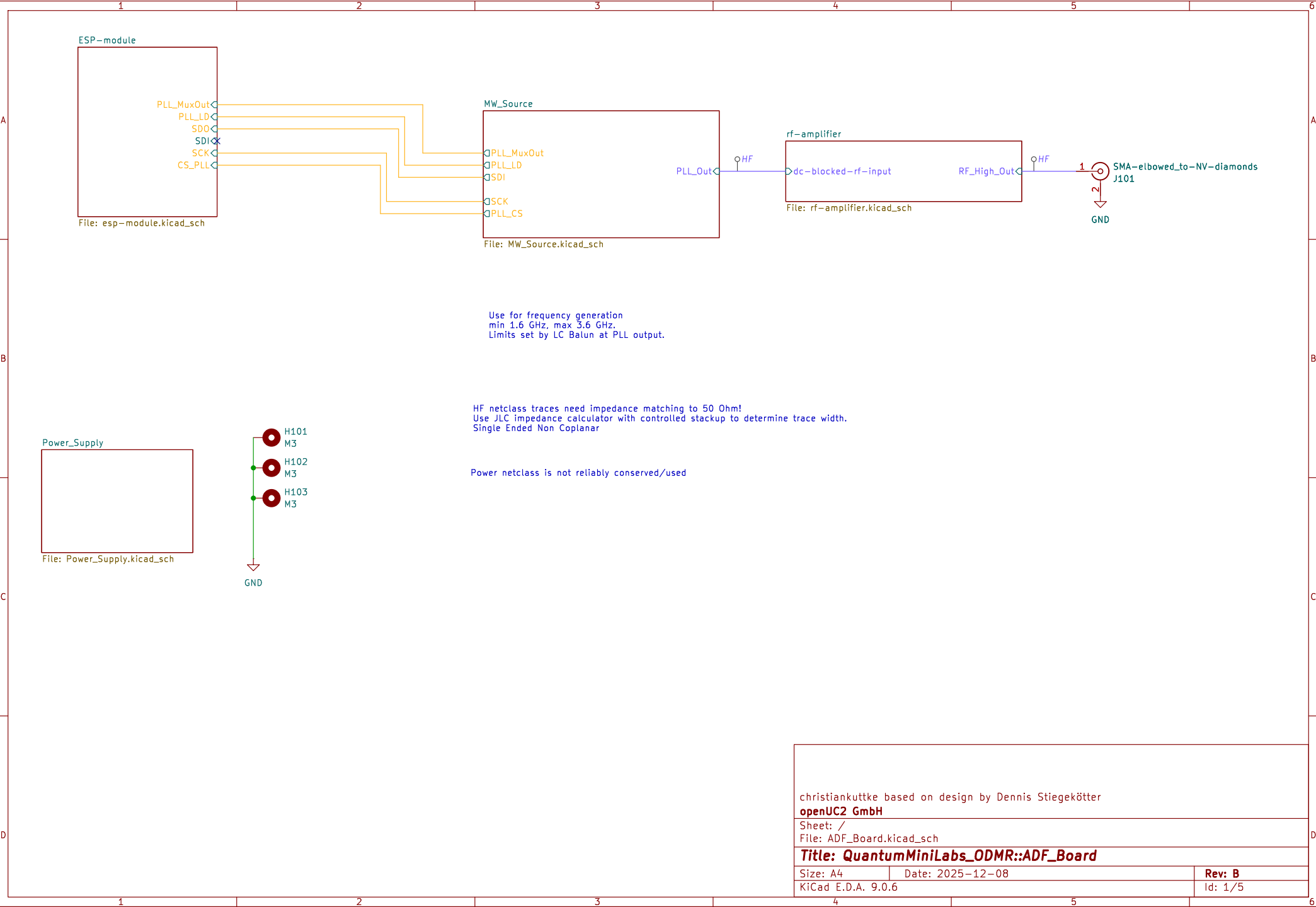
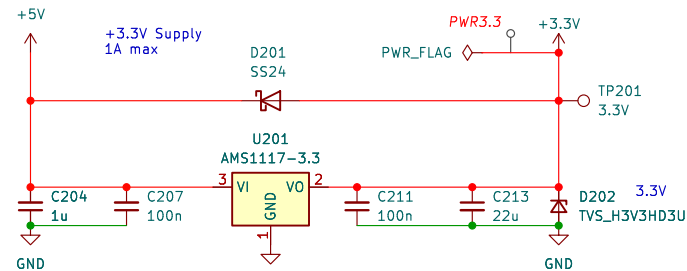
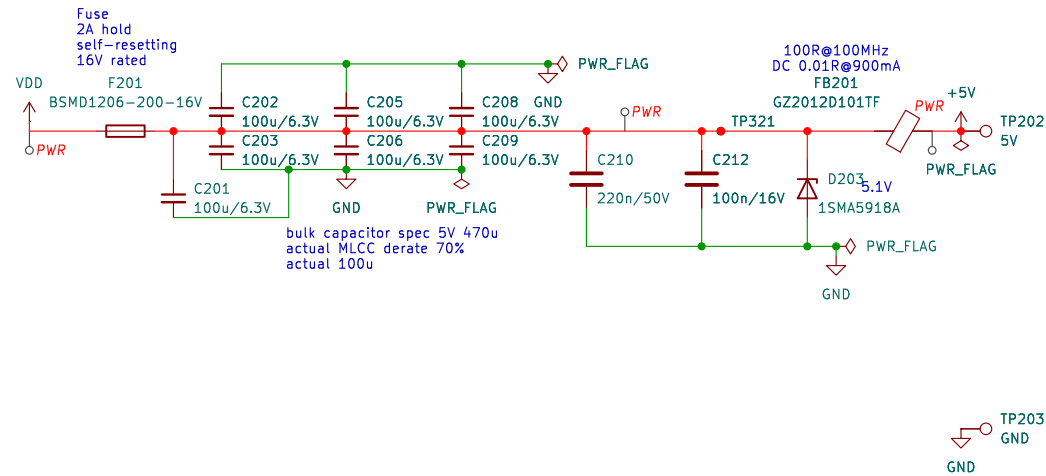




5V USB Power from XIAO Board



3,3V for PLL chip

christiankuttke based on design by Dennis Stiegekötter

openUC2 GmbH

Sheet: /Power_Supply/

File: Power_Supply.kicad_sch

Title: QuantumMiniLabs_ODMR::ADF_Board

Size: A4

Date: 2025-12-08

Rev: B

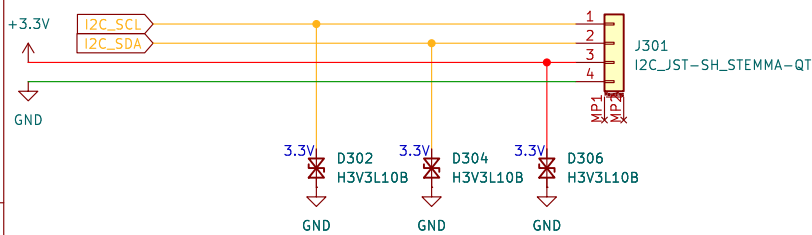
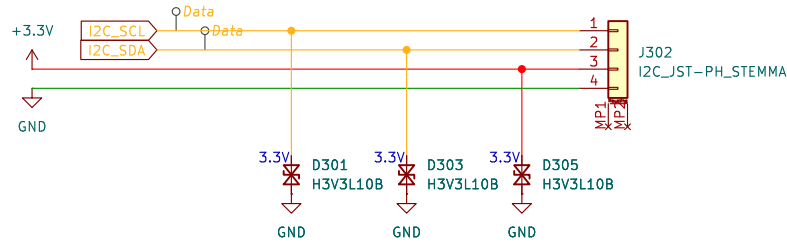
KiCad E.D.A. 9.0.6

Id: 2/5

STEMMA and STEMMA QT

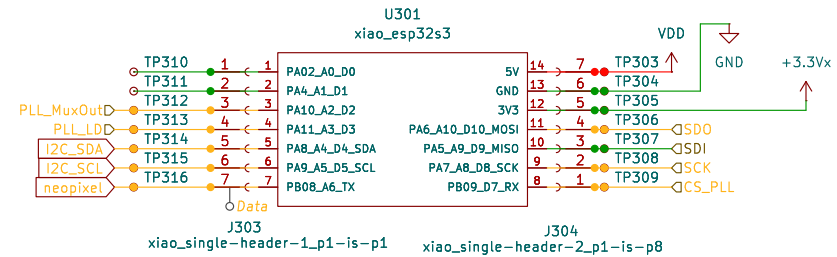
STEMMA and STEMMA QT for expansion
3.3V+I2C with max. 700 mA output at 3.3V

pinout order is correct
double checked for PH and SH



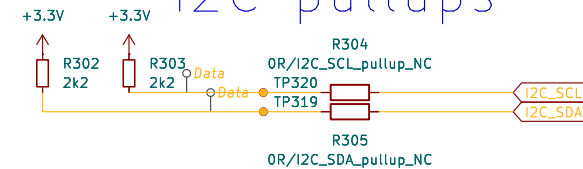
ODMR: Use with Adafruit TSL2591 module as contrast detector
featuring AMS TSL2591 light-to-digital sensor.
Adafruit TSL2591 has STEMMA QT connector
(JST SH 4-pin 1.0mm pitch).
I2C address is 0x29.

XIAO ESP32S3

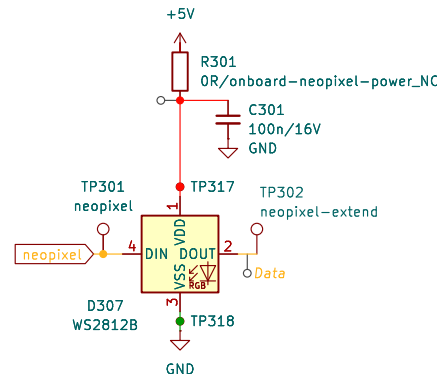


Den Mux-Ausgang der PLL habe ich nie gebraucht.
Falls noch Bedarf an freien Pins des Xiao Boards
besteht, kann die PLL_MuxOut Verbindung entfernt
werden. Auch habe ich nie Daten von dem IQ-
Modulator über SPI empfangen und die entsprech-
ende Leitung könnte bei Bedarf ebenfalls
ersetzt werden.

I2C pullups



Neopixel



christiankuttke based on design by Dennis Stiegekötter

openUC2 GmbH

Sheet: /ESP-module/

File: esp-module.kicad_sch

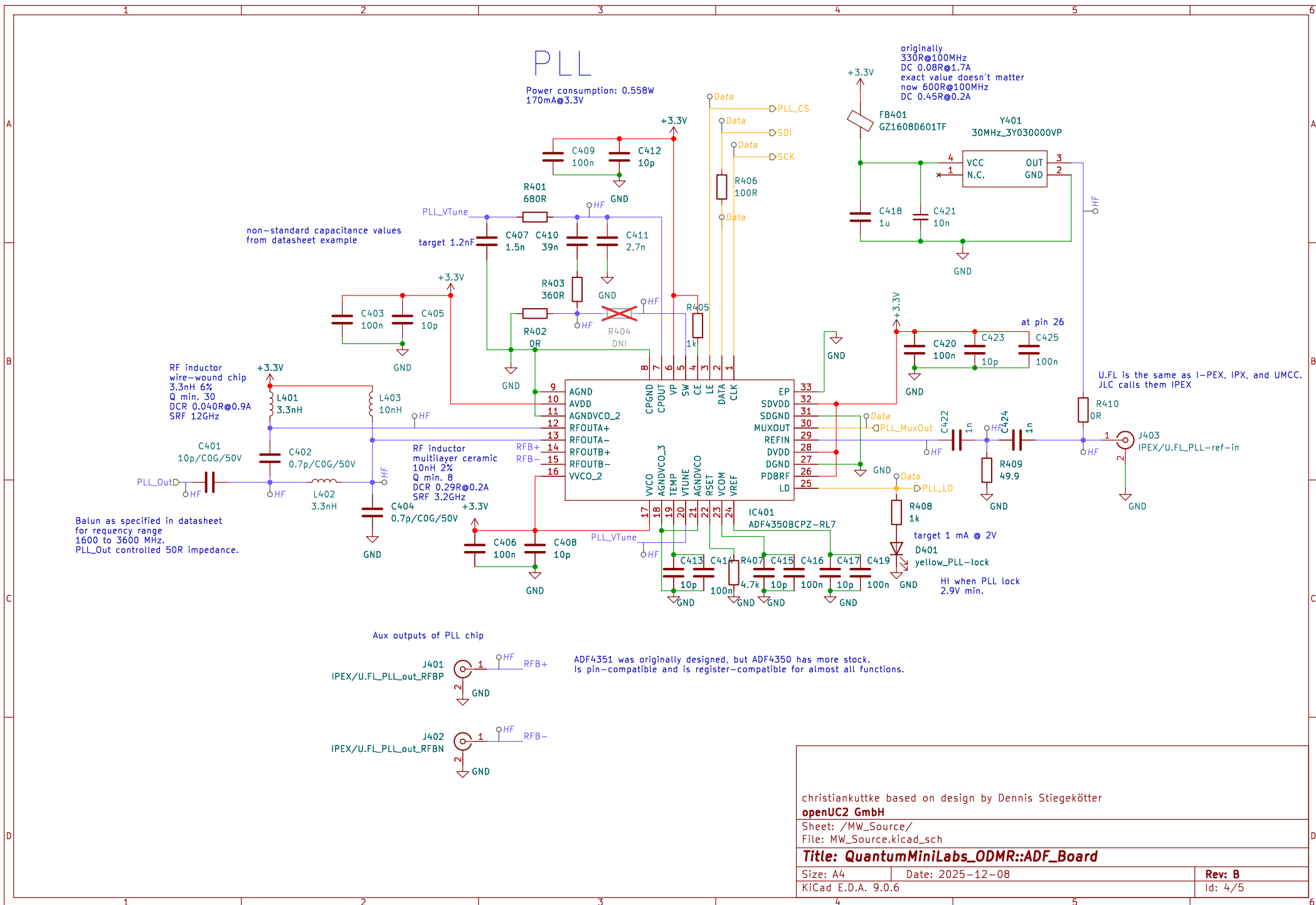
Title: QuantumMiniLabs_ODMR::ADF_Board

Size: A4 Date: 2025-12-08

KiCad E.D.A. 9.0.6

Rev: B

Id: 3/5



Gain Stage +20dB

Input DC-blocking cap required:
is at PLL output balun and
capacitance is determined by freq.
Datasheet example is 100nF (big!)

20dB gain block
50–4000MHz
5V, 106 mA max.
I/O Z matched to 50R

U501
ADL5602ARKZ-R7

+5V

C501
1u

C502
1.5n

C503
68p

GND

L501

LQW18CNR47J00D

470nH@10MHz
RMS 500mA
DCR 0.45R
SRF 555MHz

Using a wire-wound inductor way above its SRF (540MHz)
so that it filters out GHz with low DCR!

C504
10p/C0G/50V

Output DC-blocking cap:
is high-pass filter.
Higher capacitance means lower HPF cutoff.
Impedance is also a factor.
Use online DC blocking capacitor calculator.
Was at 1 nF – cutoff 3MHz
now 10 pF – cutoff 318MHz

dc-blocked-rf-input

1

GND

2

GND

3

HF

RF_High_Out

christiankuttke based on design by Dennis Stiegekötter

openUC2 GmbH

Sheet: /rf-amplifier/

File: rf-amplifier.kicad_sch

Title: QuantumMiniLabs_ODMR::ADF_Board

Size: A4

Date: 2025-12-08

Rev: B

KiCad E.D.A. 9.0.6

Id: 5/5